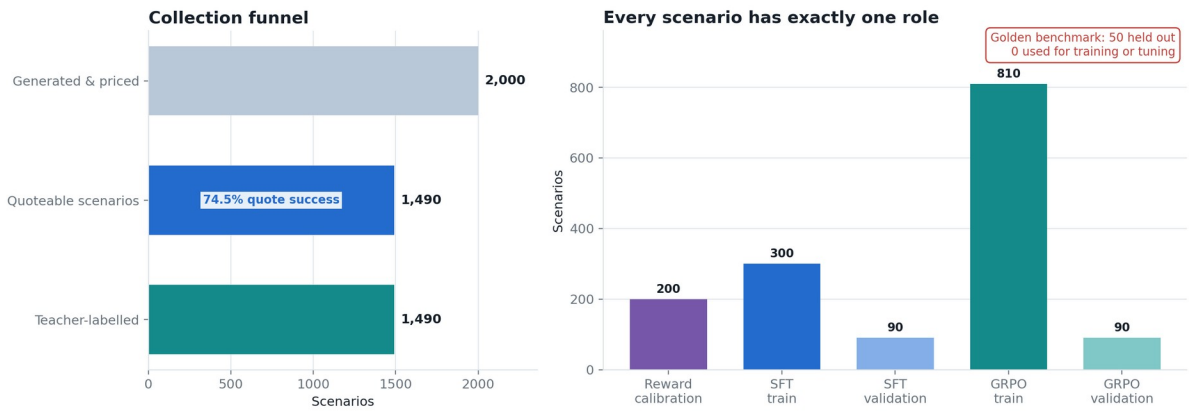


# Compact model training for insurance tariff ranking

Reaching top-tier ranking quality with Qwen3.5-2B through supervised fine-tuning, a controlled GRPO continuation, and a verified quality–cost evaluation

|                |              |                 |              |
|----------------|--------------|-----------------|--------------|
| 84%            | 86.7%        | 2B              | \$7.80       |
| held-out Top-1 | Top-3 recall | base parameters | total billed |

A deliberately separated data pipeline



Source: frozen quoted-data, split, and teacher manifests. Split counts are disjoint and exhaustive (n=1,490).

End-to-end experiment: independent data roles and a locked final benchmark

Executive conclusion

The selected SFT LoRA moved the unchanged 2B base from 22% to 84% held-out Top-1, matched the best observed frontier/teacher agreement, produced 100% strict JSON, and served at 63.66 recommendations/s in the sustained benchmark. GRPO was stable and promising on one development error, but SFT had already reached the observed supervision ceiling; ten steps did not establish a broader gain.

Report map: Task and success criteria → system and data → teacher and reward → model training → evaluation → economics → release and next experiments.

# Executive decision dashboard

| <b>Bottom line</b><br>Select Qwen3.5-2B SFT LoRA; retain the symbolic scorer as a Top-1 fallback and GRPO as a future experiment, not the production checkpoint. |                               |                               |                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------------|-----------------------------|
| Stage                                                                                                                                                            | What we did                   | Why                           | Verified result             |
| Data                                                                                                                                                             | Generated risks + live quotes | Create task-specific panels   | 1,490 usable / 9,632 quotes |
| Teacher                                                                                                                                                          | GPT-6 Sol silver labels       | Scale ranked supervision      | 1,490 Top-3 labels          |
| Reward                                                                                                                                                           | 20k deterministic search      | Align proxy to teacher        | 49% → 94% calibration Top-1 |
| SFT                                                                                                                                                              | 2B BF16 LoRA                  | Test smaller-model hypothesis | 22% → 84% golden Top-1      |
| GRPO                                                                                                                                                             | 10-step continuation          | Test reward optimization      | 84% Top-1; no broad gain    |
| Serving                                                                                                                                                          | RTX 4090 vLLM benchmark       | Measure deployability         | 63.66 rec/s at c64          |

## Decision matrix

| Option            | Top-1  | Top-3    | Runtime | Decision                            |
|-------------------|--------|----------|---------|-------------------------------------|
| Calibrated scorer | 84%    | 79.3%    | None    | Fallback when only rank one matters |
| 2B SFT            | 84%    | 86.7%    | Low     | Selected                            |
| 2B GRPO-10        | 84%    | 86.0%    | Low     | Ablation / future continuation      |
| Frontier APIs     | 80–84% | 79.3–88% | API     | Teacher/comparator, not deployment  |

## What made the result credible

- All experimental roles were disjoint; golden-risk overlap was zero.
- The canonical prompt builder controlled teacher, SFT, GRPO, and evaluation inputs.
- Reward weights were frozen before golden evaluation; the golden set was not used for tuning.
- Claims in this report are tied to machine-readable summaries, run records, and hashes.

**Git repository:** <https://github.com/lakshaychhabra/reranking-optimisation>

**Hugging Face dataset:** <https://huggingface.co/datasets/lakshaychhabra/wg-recommendation-data>

# Task and success criteria

| <b>Bottom line</b><br>Given a building risk and its priced quote panel, return three distinct offered tariffs in ranked JSON; the primary target was held-out Top-1 agreement above 80%. |                                                      |                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------|
| Metric                                                                                                                                                                                   | Definition                                           | Why it matters                                   |
| Top-1 agreement                                                                                                                                                                          | First recommendation equals reference rank one       | Primary assignment measure                       |
| Top-3 recall                                                                                                                                                                             | Reference tariffs recovered across three slots       | Measures shortlist usefulness                    |
| Strict JSON                                                                                                                                                                              | Exact required schema                                | Operational reliability                          |
| Reward ratio                                                                                                                                                                             | Supplied reward of model rank one / scenario maximum | Audits quality-price trade-off; not ground truth |
| Serving cost                                                                                                                                                                             | Measured GPU cost per 1,000                          | Deployment economics                             |

## Reference hierarchy

Fable rankings are the held-out evaluation reference. GPT-6 Sol is scalable silver supervision. The calibrated numeric reward is a deterministic approximation to that teacher. The objects are related, but none should be presented as objective truth.

**Important correction**  
The supplied 9B model scored 42%. Our experiment did not fine-tune that model: our unchanged 2B base scored 22%, and our 2B SFT checkpoint scored 84%. Therefore the training gain is 0.22 → 0.84; 0.42 is a separate assignment reference.

## Output contract

`{"top_3":[{"rank":1,"tariff_id":"49"}, {"rank":2,"tariff_id":"48"}, {"rank":3,"tariff_id":"162"}]}`

# System and experimental pipeline

**Bottom line**

One canonical prompt and explicit data roles connected scenario generation, real quote collection, teacher labels, reward calibration, SFT, GRPO, and evaluation.

A deliberately separated data pipeline



Source: frozen quoted-data, split, and teacher manifests. Split counts are disjoint and exhaustive (n=1,490).

Dataset funnel and experimental separation

| Control             | Implementation                        | Risk controlled           |
|---------------------|---------------------------------------|---------------------------|
| Scenario identity   | Hash of normalized risk JSON          | Duplicate leakage         |
| Quote access        | Cached client; ≥3 s live-call spacing | Cost and rate safety      |
| Prompt construction | src/prompt.py only                    | Training/evaluation drift |
| Quote order         | Seeded shuffle with stored mapping    | Positional bias           |
| Golden boundary     | Final evaluation only                 | Selection leakage         |

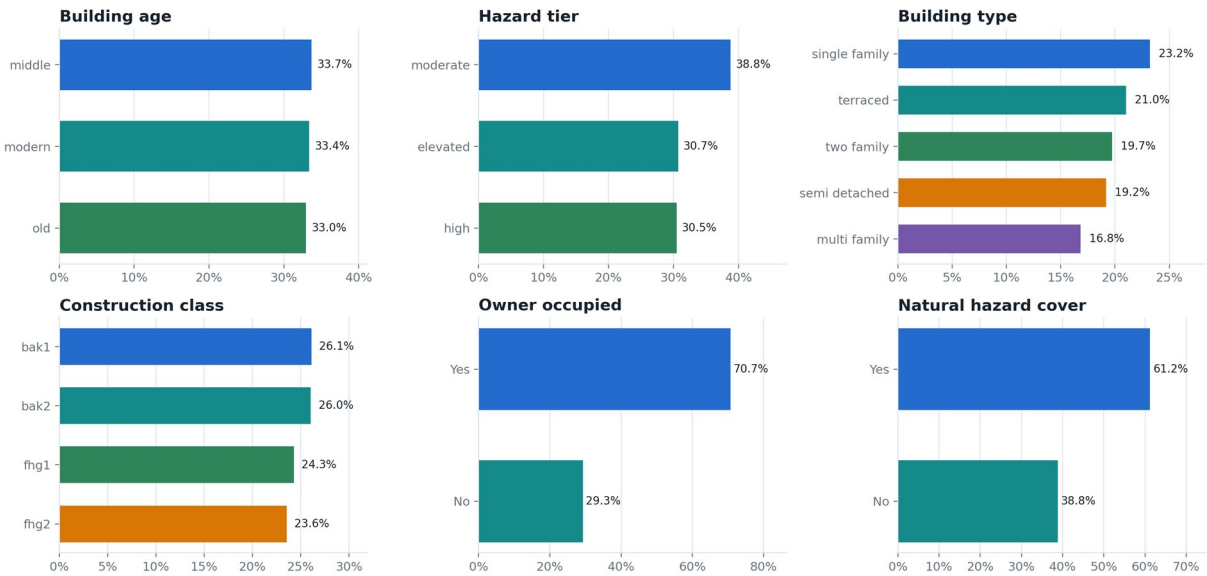
## Why this architecture

The design separates four questions: Can we create representative quote panels? Can a teacher provide consistent rankings? Can a deterministic reward capture the teacher’s preference? Can a small learned policy generalize beyond that formula? This makes failures attributable rather than mixing data, objective, and optimization effects.

# Dataset creation and coverage

**Bottom line**  
The final corpus contains 1,490 usable scenarios and 9,632 live-engine quotes, then assigns every scenario exactly one experimental role.

**The synthetic portfolio is broad—and the splits preserve it**



Source: split manifest. n=1,490 scenarios, 9,632 quotes, 5 insurers, 9 products. Largest final-split marginal deviation: 1.56 pp.

**Scenario and quote-panel coverage in the generated corpus**

| Split                   | N        | Labels               | Purpose                              |
|-------------------------|----------|----------------------|--------------------------------------|
| Reward calibration      | 200      | Teacher              | Reward selection and validation      |
| SFT train / validation  | 300 / 90 | Teacher              | Imitation and checkpoint selection   |
| GRPO train / validation | 810 / 90 | Train labels omitted | Policy optimization and offline gate |
| Golden                  | 50       | Fable                | Final reporting only                 |

## Checks

- 1,490 assignments were disjoint and exhaustive.
- No generated risk exactly overlapped a golden risk.
- Final panels span five insurers and nine tariff IDs.
- Only panels with at least three offered quotes entered training.

Scope note: generated risks and the finite product catalogue support a controlled experiment, not a claim of complete production-population coverage.

# Teacher model and supervision

| <b>Bottom line</b><br>GPT-6 Sol produced 1,490 ranked Top-3 labels for \$3.64 in the batch run; including teacher selection, the label programme cost approximately \$3.80. |                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Item                                                                                                                                                                        | Verified value      |
| Teacher                                                                                                                                                                     | GPT-6 Sol           |
| Annotations                                                                                                                                                                 | 1,490               |
| Input / output tokens                                                                                                                                                       | 1,546,926 / 417,797 |
| Batch annotation cost                                                                                                                                                       | \$3.635911          |
| Teacher selection + labels                                                                                                                                                  | ≈\$3.80             |
| Observed golden Top-1                                                                                                                                                       | 84% (42/50)         |

## Why use a teacher

Direct Fable labels were available only for the locked 50-scenario benchmark. A frontier teacher made task-specific supervision affordable at 1,490-scenario scale. The same canonical prompt and quote-ID mapping were used throughout, and the raw numeric reward was not exposed to the teacher.

## What the 84% teacher result means

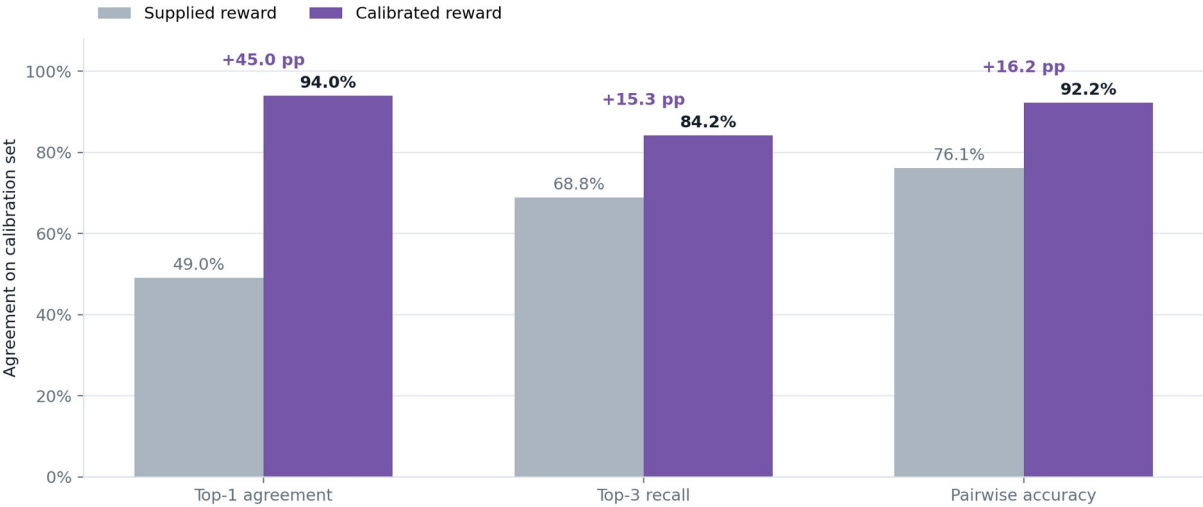
It is an observed ceiling for this supervision pipeline, not a theorem about all possible models. SFT matching 84% indicates that additional optimization against the same teacher/reward proxy had limited demonstrated headroom against Fable. Better expert labels could change that ceiling.

|                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Economic motivation</b><br>A frontier/Fable-quality API route, represented by the evaluated Gemini and GPT-5.4 comparators, was materially more expensive than the supplied 9B reference: about 5× and 7× its per-recommendation cost. This is a comparator-based estimate, not a measured Fable tariff. Teacher-supervised fine-tuning aimed to retain that ranking quality in a self-hostable 2B model. |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

# Reward grid search and validation

**Bottom line**  
Reward calibration showed that the effective task was a quality-price trade-off; a frozen deterministic scorer matched 84% Fable Top-1 but remained weaker on the full Top-3.

## Reward calibration makes the scorer look more like the teacher



Source: 200-scenario reward-calibration split only; 5-fold selection, no golden or GRPO-validation data. Not directly comparable to golden metrics.

### Twenty-thousand-candidate search and held-out controls

| Evaluation               | Default | Frozen calibrated |
|--------------------------|---------|-------------------|
| Calibration Top-1        | 49%     | 94%               |
| Calibration Top-3 recall | 68.8%   | 84.2%             |
| Golden Top-1             | 38%     | 84%               |
| Golden Top-3 recall      | 65.3%   | 79.3%             |
| Golden pairwise          | 66.0%   | 90.4%             |

**Frozen formula**  
$$\text{reward} = (0.2863 + 0.4273 \times \text{score}_n) / (1 + 1.7936 \times \text{price}_n)$$

The selected weights were unchanged during validation; calibration used no golden labels and no LLM calls. A permutation negative control returned chance-level pairwise ranking, reducing the risk that the search merely manufactured agreement.

# Model selection and training design

| <b>Bottom line</b><br>Qwen3.5-2B was chosen to challenge the assumption that a 9B model was necessary; BF16 LoRA kept the experiment transparent and operationally simple. |                                |                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-----------------------------------------------------------------------|
| Choice                                                                                                                                                                     | Selected                       | Reason                                                                |
| Base model                                                                                                                                                                 | Qwen3.5-2B                     | Small enough to test the capacity hypothesis; modern instruction base |
| Adaptation                                                                                                                                                                 | BF16 LoRA                      | Simple pipeline; no quantization confound; fits available GPU         |
| LoRA                                                                                                                                                                       | rank 16 / alpha 16 / dropout 0 | 10.9M trainable parameters (0.49%)                                    |
| SFT data                                                                                                                                                                   | 300 train / 90 validation      | Independent teacher-labelled pool                                     |
| SFT schedule                                                                                                                                                               | 3 epochs / 21 steps            | Validation-loss checkpoint selection                                  |
| Loss                                                                                                                                                                       | Completion only                | Learn output policy without fitting prompt tokens                     |

## Why not QLoRA

The full BF16 base plus LoRA fit the selected hardware, so quantization was not required. Avoiding QLoRA kept the experiment simpler and removed another variable from a short, decision-oriented comparison.

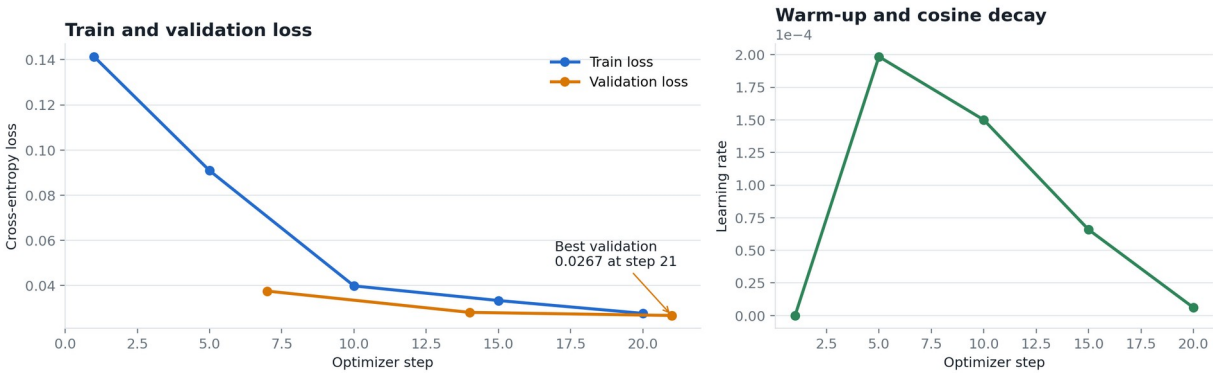
## What would change the model choice

If time allowed, the next capacity experiment would test  $\leq 1$ B models under the same split and evaluation contract. The 2B result supports—not exhausts—the hypothesis that this structured task needs far less than 9B parameters.

# Supervised fine-tuning

**Bottom line**  
SFT transformed both ranking quality and operational reliability: 22% → 84% Top-1 and 2% → 100% strict JSON.

**SFT converges quickly on a small, clean target**



Source: frozen SFT run record. 300 train / 90 validation examples; 3 epochs; 6.3 min elapsed; 12.6 GB peak GPU memory.

SFT learning curve and selected checkpoint

| Golden metric          | 2B base | 2B SFT | Change     |
|------------------------|---------|--------|------------|
| Top-1                  | 22%     | 84%    | +62 pp     |
| Top-3 recall           | 66.7%   | 86.7%  | +20 pp     |
| Strict JSON            | 2%      | 100%   | +98 pp     |
| Valid three unique IDs | 100%    | 100%   | Maintained |

## Interpretation

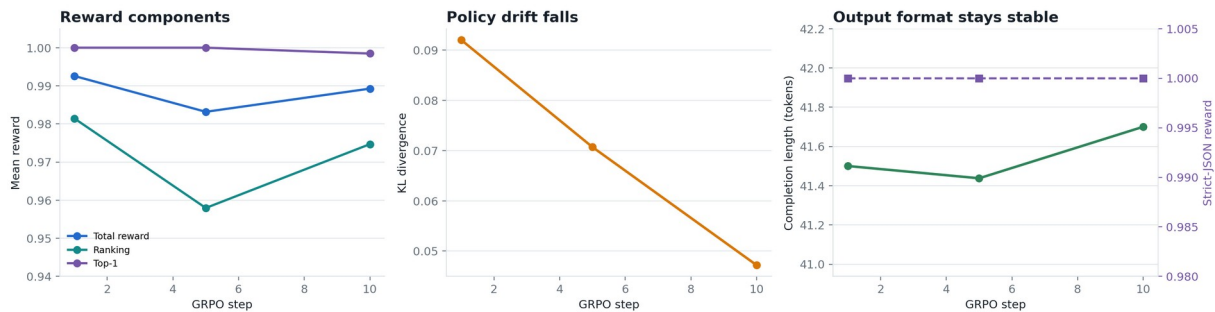
The improvement is not merely format compliance: Top-1 and Top-3 both rose sharply. The selected model’s lower supplied-reward ratio than price-dominant baselines reflects a deliberate shift toward the teacher/Fable quality preference, not an internal contradiction.

**Selection rule**  
The SFT checkpoint was selected from validation loss on its own 90-scenario split. Golden results were reported afterward and were not used to choose SFT steps or hyperparameters.

# GRPO continuation: useful signal, limited evidence

**Bottom line**  
GRPO did not fail: it was stable and fixed one development Top-1 error. Ten steps were insufficient to demonstrate a broad improvement beyond an SFT model already at the teacher’s observed Fable agreement.

**The 10-step GRPO continuation is stable—but nearly saturated**



Source: frozen GRPO run record. 810 train / 90 validation examples; 10 optimizer steps; 5.3 min elapsed; 6.9 GB peak GPU memory. Only three logged train observations—read as diagnostics, not a smooth trend.

**Ten-step GRPO training diagnostics**

| Evaluation               | SFT   | GRPO-10 | Observed change       |
|--------------------------|-------|---------|-----------------------|
| Development Top-1        | 83/90 | 84/90   | +1 corrected scenario |
| Development Top-3 recall | 87.0% | 86.3%   | Small net loss        |
| Golden Top-1             | 42/50 | 42/50   | No change             |
| Golden Top-3 recall      | 86.7% | 86.0%   | One fewer hit / 150   |
| Strict JSON              | 100%  | 100%    | No degradation        |

## Why stop

- The assignment’s >80% objective was already exceeded by SFT.
- SFT matched the teacher’s observed 84% Fable Top-1.
- Additional optimization would still target a silver reward proxy, not independent truth.
- The time budget favored verifying reproducibility and serving economics over an open-ended RL sweep.

Longer training, more samples per prompt, or better supervision could still correct remaining cases. The defensible claim is that ten steps did not establish a broad gain—not that GRPO cannot work.

# Final evaluation and model selection

| <b>Bottom line</b><br>The selected 2B SFT model matched the best observed Top-1 result and produced the strongest balanced Top-3 among the compact deployable options. |       |              |              |                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------|--------------|--------------------|
| System                                                                                                                                                                 | Top-1 | Top-3 recall | Reward ratio | Cost / 1k          |
| Supplied Qwen3.5-9B                                                                                                                                                    | 42%   | 66.0%        | 0.983        | €0.1869            |
| Our Qwen3.5-2B base                                                                                                                                                    | 22%   | 66.7%        | 0.844        | —                  |
| Calibrated scorer                                                                                                                                                      | 84%   | 79.3%        | 0.812        | €0 inference       |
| Our 2B SFT — selected                                                                                                                                                  | 84%   | 86.7%        | 0.812        | €0.0028–0.0539     |
| Our 2B GRPO-10                                                                                                                                                         | 84%   | 86.0%        | 0.812        | Same serving class |
| Gemini                                                                                                                                                                 | 80%   | 88.0%        | 0.763        | €0.8897            |
| GPT-5.4                                                                                                                                                                | 84%   | 79.3%        | 0.802        | €1.3445            |
| GPT-6 Sol                                                                                                                                                              | 84%   | 86.7%        | 0.832        | €2.8707            |

## Selection rationale

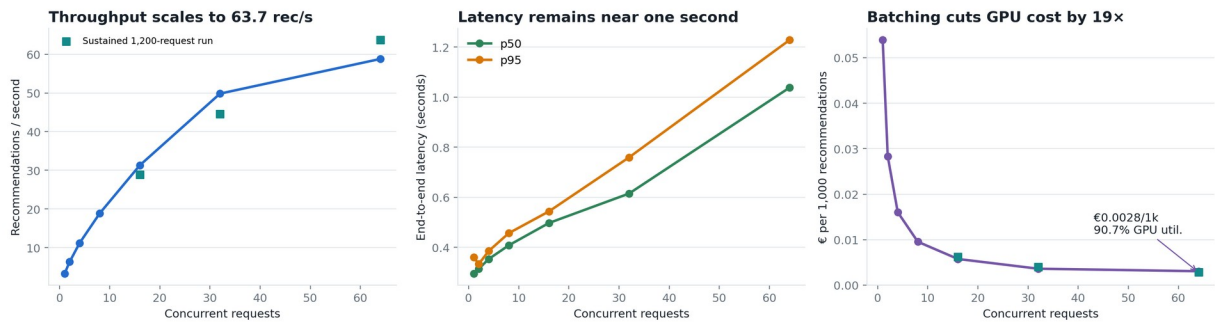
SFT ties the strongest Top-1, matches GPT-6 Sol Top-3, exceeds the symbolic scorer by 11 recovered Top-3 positions, satisfies the output contract, and has a measured low-cost serving path. GRPO adds complexity without a demonstrated aggregate benefit at the tested budget.

The calibrated scorer remains valuable: if a product consumes only rank one, it offers the same measured Top-1 with no inference. It is not chosen for this deliverable because the required output is a useful ranked Top-3.

# Serving benchmark and economics

**Bottom line**  
A dedicated post-fix vLLM benchmark—not incidental training-run timing—provides the serving figures used for the cost claim.

**Continuous batching turns a small model into cheap capacity**



Measured with the selected Qwen3.5-2B SFT adapter, vLLM 0.29.0, BF16, and ~1,065 tokens/recommendation on a RunPod RTX 4090 (\$0.74/h). Squares are sustained 1,200-request confirmations; all requests produced valid top-3 JSON.

**Measured throughput, latency, and utilization on an RTX 4090**

| Scenario                  | Cost / 1k | Status                         |
|---------------------------|-----------|--------------------------------|
| Saturated, concurrency 64 | €0.002832 | Measured; 63.66 rec/s          |
| 50% utilization           | €0.005663 | Scenario                       |
| 25% utilization           | €0.011327 | Scenario—not a recommendation  |
| 10% utilization           | €0.028317 | Scenario                       |
| Unbatched / concurrency 1 | €0.053899 | Measured lower-throughput mode |

## Timing caveat

Early end-to-end runs took longer because FlashAttention was not functioning correctly; that environment issue was later fixed. For this reason, the report does not infer production latency from training or ad-hoc evaluation timestamps. It uses the separate serving benchmark: 1,200 requests, p50 1.003 s, p95 1.105 s, 90.7% mean GPU utilization, and 100% valid outputs.

## Cost boundaries

Serving estimates use a \$0.74/hour RTX 4090 and are GPU-only. Storage, networking, cold starts, redundancy, monitoring, and engineering operations are excluded.

## Spend, reproducibility, and delivery

### Bottom line

The complete billed experiment cost approximately \$7.80, and the release package preserves adapters, configs, logs, frozen outputs, hashes, and a portable reproduction flow.

| Spend item                | Amount     | Evidence                   |
|---------------------------|------------|----------------------------|
| Teacher pilot / selection | ≈\$0.16    | Selection records          |
| Teacher batch labels      | \$3.635911 | annotation_summary.json    |
| RunPod total              | \$4.00     | Owner-confirmed total bill |
| Total                     | ≈\$7.80    | Experiment + benchmarking  |

### Release contents

- SFT and GRPO adapters with hashes and resolved configs.
- Frozen base, SFT, GRPO, comparator, and serving summaries.
- Prepared non-golden data and split manifests.
- HOW\_TO\_RUN.md and portable reproduction script.
- Apple MPS verification: all 50 SFT raw outputs matched CUDA exactly.

**Git repository:** <https://github.com/lakshaychhabra/reranking-optimisation>

**Hugging Face dataset:** <https://huggingface.co/datasets/lakshaychhabra/wg-recommendation-data>

### Reproduction rule

Use development data for routine checks. The golden benchmark remains final-report evidence and must not become an iterative checkpoint-selection loop.

# Limits and next experiments

| <b>Bottom line</b><br>The current evidence supports the assignment decision; production confidence requires stronger independent labels, more scenarios, and a capacity study below 2B. |                                           |                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-----------------------------------------------------------------|
| Limitation                                                                                                                                                                              | Why it matters                            | Next experiment                                                 |
| 50-scenario golden set                                                                                                                                                                  | One case moves Top-1 by 2 pp              | Larger untouched expert-labelled benchmark                      |
| Silver supervision                                                                                                                                                                      | Teacher can reproduce its own preferences | Expert adjudication on disagreements                            |
| Synthetic risks / finite catalogue                                                                                                                                                      | May miss operational tail cases           | Shadow evaluation on production-like traffic                    |
| Fixed reward formula                                                                                                                                                                    | Can miss feature interactions             | Compare learned reward or preference model                      |
| 2B is smallest tested                                                                                                                                                                   | Capacity floor remains unknown            | Repeat SFT at $\leq 1B$                                         |
| GRPO only 10 steps                                                                                                                                                                      | Longer run may improve remaining errors   | Predeclare dev gate; increase steps/samples after better labels |

## Recommended sequence

- Deploy or shadow-test the selected SFT adapter with the symbolic scorer as a monitoring baseline.
- Collect expert labels specifically where SFT, scorer, and teacher disagree.
- Run the  $\leq 1B$  capacity ablation under the identical prompt and split contract.
- Only then extend GRPO with a frozen development rule and a new untouched final benchmark.

|                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Final conclusion</b><br>The assignment aim was achieved: a small self-hostable model exceeded 80% Top-1 and matched the observed top-tier result at a fraction of serving cost. The main lesson is not that one optimizer won; it is that disciplined data separation, reward interpretation, and simple SFT captured nearly all demonstrated value, while GRPO remains a credible path once the supervision ceiling is raised. |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|